

LAB # 9

IF THEN ELSE, CONTROL STRUCTURE IN MIPS

OBJECTIVE

To implement translation of “if then else” control structures in mips assembly language.

Theory

Translation of an “IF THEN ELSE” Control Structure

The “If (condition) then do {this block of code} else do {that block of code}” control structure is probably the most widely used by programmers. Let us suppose that a programmer initially developed an algorithm containing the following pseudocode.

```
if($t8 < 0)
{$s0 = 0 - $t8;
$t1 = $t1 + 1}
else {$s0 =
$t8;
$t2 = $t2 + 1}
```

When the time comes to translate this pseudo code to MIPS assembly language the results could appear as shown below. In MIPS assembly language, anything on a line following the number sign (#) is a comment. Notice how the comments in the code below help to make the connection back to the original pseudocode.

sub \$s0, \$zero, \$t8 # \$s0 gets the negative of \$t8	bgez \$t8, else # if (\$t8 is > or = zero) branch to else
addi \$t1, \$t1, 1 # increment \$t1 by 1	
b next # branch around the else code	
else:	
ori \$s0, \$t8, 0 # \$s0 gets a copy of \$t8	
addi \$t2, \$t2, 1 # increment \$t2 by 1	
next:	

BRANCH IF GREATER THAN ZERO

```
bgez $v0,else # branch to else if $v0>=0
```

SHIFT LEFT LOGICAL

```
sll $t2,$v0,2 # set $t2 to result of shifting $v0 left by number specified by  
               immediate
```

SHIFT RIGHT LOGICAL

```
srl $t2,$v0,2 # set $t2 to result of shifting $v0 right by number specified immediate
```

PROGRAM#1:

Write a program in MIPS assembly language that translate IF-THEN ELSE control structure.

```
##### DATA SEGMENT #####
```

```
.data    input: .asciiz "\n type any  
number"  rshift: .asciiz "\n number after  
right shift: "  lshift: .asciiz "\n number  
after left shift: "
```

```
##### TEXT SEGMENT #####
```

```
.text  
.globl main  
main: la $a0,input #  
print input message  li $v0,4  
syscall li $v0,5      syscall  
bgez $v0,else # if- else equivalent statement  
srl $t2,$v0,2 # shift right logical la $a0,rshift li  
$v0,4 syscall # print value of move  
$a0,$t2 li $v0,1 syscall
```

```
b end #branch to statement at end unconditionally else:
```

```

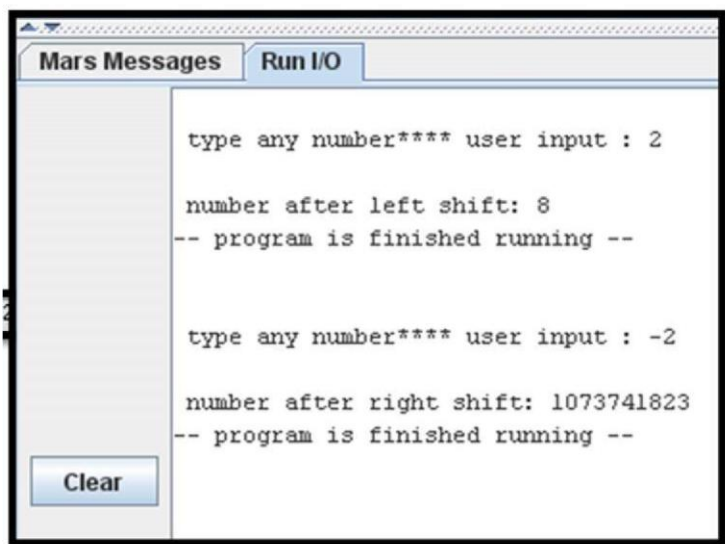
sll $t2,$v0,2 #shift left logical la
$a0,$t2 # print value of $t2 li
$v0,4 syscall

move $a0,$t2 # print value after left shift
li $v0,1 syscall

end: li $v0,10 # terminate
program
syscall

```

OUTPUT:



LAB TASKS:

1. Write a program in MIPS assembly language that take input from user and print whether input is greater or less than 10 and also shift input left and right 4 bits.
2. Write a program in MIPS assembly language that takes an integer input from the user, checks whether the number is even or odd. If the number is even, the program performs a left shift by 2 bits, otherwise it performs a right shift by 1 bit, and then displays the result